

HSS201 History of Science
End-Semester Examination

Time: 2 hrs

Total Marks: 50

Answer **Any 25** from the following:

1. Which of the following is known as the *Florentine Codex*, a monumental work authored by Bernardino de Sahagun that described, not only the plants and animals of the New World, but also Aztec medicine, religion and history?

- (a) Botanical Specimens of Mexico (c) **General History of the Things of New Spain**
(b) Fossil Records of the New World (d) Natural and Moral History of the Indies

2. Francisco Hernández returned to Spain from the New World with sixteen handwritten volumes along with paintings, later published as *The Treasury of Medical Matters of New Spain* (1628). This work, among others, established that -

- (a) **Knowledge originating in the New World, produced by the Indigenous people of the Americas, helped shape a vision of medicine as an experimental and practical science**
(b) Superiority of European knowledge systems
(c) Inferiority of Indigenous knowledge systems
(d) Adventurous nature of European travellers

3. Emperor Moctezuma II (1466-1520 CE) who was ruling the Aztec empire in Mexico from the capital city of Tenochtitlan maintained -

- (a) A strong army and advanced weaponry (c) Advanced naval forces
(b) **Aviary, Botanical Garden and Menagerie** (d) Numismatic Mint

4. One of the important streams of thought in the Scientific Revolution -

- (a) Theology (c) Oriental Philosophy
(b) Conformism (d) **Mechanical Philosophy**

5. The Anglo-Irish natural philosopher Robert Boyle (1627-1691) argued that air was made up of

- (a) **Spring like particles** (c) Gas
(b) Vacuum (d) Liquids

6. Many natural philosophers argued that nature can be understood mathematically. This represented a major shift in the epistemological – and social – status of mathematics. The result was that Mathematicians were

- (a) Suspected (c) **Given a position in princely courts and aristocratic households**
(b) Punished (d) Banished

7. Inspired by the Weberian sociology, Robert K Merton, in his *Science, Technology, and Society in Seventeenth-Century England*, investigated the relationship between -

- (a) Greek philosophy and modern science (c) Finance capitalism & science
(b) Christianity and Greek Philosophy (d) **ascetic Protestantism and scientific inquiry**

8. The Needhamian approach to the history of science entails consideration of science as
 (a) Western science (c) Eastern science
(b) Ecumenical science (d) Spiritual science
9. According to Joseph Needham, what are the factors that impeded the growth of science in China?
(a) State bureaucracy & low competition (c) Lack of scientific knowledge
 (b) Abundant Mathematical rationality (d) Persistence of slavery
10. The idea of 'The Scientific Revolution' that took place in the seventeenth century is a
 (a) 17th century perception (c) Perception of the Catholic Church
(b) Recent invention (d) Perception of Absolute Monarchs
11. Who posed the following question: Why did the West, rather than other civilizations develop modern capitalism, with its rational and systematic approach to economic activity?
 (a) Joseph Needham (c) Stephen Jay Gould
 (b) Robert Merton **(d) Max Weber**
12. In which of the following works did Galileo defend the Copernican model based on his observations from the telescope?
(a) Dialogues Concerning the Two Chief World Systems (c) Principia Mathematica
 (b) New Theory about Light and Colours (d) Opticks
13. During the seventeenth century, magic seemed to some natural philosophers to be a promising way of investigating nature. Magic was regarded as -
(a) a search for the hidden 'occult' qualities of natural objects & phenomena
 (b) a talent worthy of practice (c) a promising experiment (d) a Mathematical calculation
14. The Danish astronomer Tycho Brahe built his astronomical observatory at -
 (a) Paris **(c) Uraniborg**
 (b) Copenhagen (d) Oslo
15. Who invented telescope
 (a) Galileo Galilei **(c) Hans Lipperhey**
 (b) Johannes Kepler (d) Nicolaus Copernicus
16. What was NOT discovered by Galileo Galilei
 (a) Moon's of Jupiter (c) Craters on the Moon
 (b) Phases of Venus **(d) Neptune**
17. From the observation of light spots in the dark region & dark spots in the lit region of the Moon Galileo concluded they are
(a) Mountain peaks on the Moon (c) Shallow craters on the Moon
 (b) Deep holes on the Moon (d) Was a puzzle he could not answer

18. Which is NOT a book written by Galileo

- (a) Sidereus Nuncius (The Sidereal Messenger)
- (b) Philosophiæ Naturalis Principia Mathematica**
- (c) Dialogue Concerning the Two Chief World Systems
- (d) Discourses on Two New Sciences

19. Discovery of Moon's of Jupiter according to Galileo implied

- (a) There are life beyond Earth
- (c) Not everything in the sky is revolving around the Earth**
- (b) Jupiter is a large planet
- (d) Universe is infinite

20. The phases of Venus observed by Galileo implied

- (a) Venus is orbiting the Sun**
- (c) Venus is orbiting in epicycles around the Earth
- (b) Venus is orbiting the Jupiter
- (d) Venus is the hottest planet

21. Gaston Bachelard introduced the notion of

- (a) Innovation
- (c) Research programme (or Research paradigm)
- (b) Epistemological rupture (or epistemological break)**
- (d) Naturalisation

22. Epistemological rupture (or epistemological break) means

- (a) a radical, discontinuous break in a system of knowledge or understanding, where a new conceptualisation replaces an old one, rendering the previous framework obsolete**
- (b) practical implementation of ideas that result in the introduction of new goods or services or improvement in offering goods or services.
- (c) a series of interconnected studies or research activities aimed at a long-term, overarching goal
- (d) human condition where lives being shaped by natural forces and a low level of technology

23. Life of Galileo is a play written by

- (a) Samuel Beckett
- (c) Sheila Callaghan
- (b) Albert Camus
- (d) Bertolt Brecht**

24. Which is NOT Galilio's work

- (a) Experiments on inclined plane
- (c) Heliocentrism
- (b) Universal law of Gravity**
- (d) Development of pendulum clock

25. The author of the famous essay on history of science "Social and economic roots of Newton's Principia is

- (a) Joseph Needham
- (c) JD Bernal
- (b) Boris Hessen**
- (d) Charls Singar

26. ibn al-Haytham's experiments with pin-hole camera showed that

- (a) light is a wave
- (c) light can be thought as consisting of 'light rays'**
- (b) quantum nature of light
- (d) light is like scent, it wafts through the atmosphere

27. According to Ibn al-Haytham visual perception occurs due to
(a) Light rays emanating from an object strike our eyes, and like pinhole camera produce the image
 (b) Visual rays emanate from eyes and interact with objects
 (c) Visual rays from eyes and the object rays from the objects interact and produce image
 (d) Light rays are produced from eye and visual perceptions is formed
28. In early modern Europe pin-hole camera was used by
 (a) sailors to see the distant planets (c) physiologist to see internal organs
(b) artists to make realist paintings (d) mathematicians to explore geometrical forms
29. Ibn al-Haytham showed that
 (a) Moon reflects sunlight like a plane mirror into parallel rays
 (b) Moon reflects sunlight like a curved mirror into rays travelling towards earth
(c) Moon reflects light a rough surface into diffused rays
 (d) Moon is a source of light
30. In Kitāb al-Manāẓir, the word i'tibār "is exclusively applied to
(a) a procedure of controlled observations that is distinct from mere repetition of observations
 (b) a procedure of meticulous repetition of observations
 (c) a procedure of postulating hypothesis
 (d) a procedure of cumulative experience
31. Arabic Mediaeval Renaissance was inspired by
 (a) Talibanism (c) Greek orthodox Christianity
 (b) Sufism **(d) Mu'tazilism**
32. Frontispiece to Hevelius' Selenographia (1647) associated Ibn al-Haytham with
 (a) Religion **(c) Rationality**
 (b) arts (d) History
33. Galileo's daughter who helped him during his house arrest is
(a) Maria Celeste (c) Maria Francesca
 (b) Maria Allegra (d) Maria Bianca
34. Aristotle classified motion fundamentally into
 (a) Natural and Circular (c) Natural and Elliptical
 (b) Natural and Irregular **(d) Natural and Violent**
35. Johannes Kepler saw a 'new' star on October 9 1604. The next day, October 10, 1604, Galileo too observed it through a telescope. They showed that the 'new star' or the Nova of 1604 was situated beyond Moon using
 (a) Telescope **(c) Parallax**

- (b) Colour (d) Brightness

36. Stephen Jay Gould argued in his *Time's Arrow, Time's Cycle: Myth and Metaphor in the Discovery of Geological Time* that the period from 1750 to 1900 is characterized by -

- (a) the rise of theological imagination (c) **the ascent of historical thinking**
(b) the rise of Oriental imagination (d) the descent of metaphysical thinking

37. According to Hegel's history of philosophy, the world spirit's march is towards -

- (a) **Absolute freedom** (c) Absolute democracy
(b) Absolute slavery (d) Absolute oligarchy

38. Who propounded the following thesis: During the early stages of institutionalization the legitimacy of western science was an unintended consequence of values and practices of ascetic Protestantism. Among these scientists one could count English Protestants and German Pietists:

- (a) Max Weber (c) **Robert K. Merton**
(b) Joseph Needham (d) Karl Marx

39. One of the limits of Max Weber's understanding of India is -

- (a) Empiricism (c) He relied on ancient sources
(b) He visited India briefly (d) **He possessed limited knowledge of social history**

40. Who authored *The Two Cultures and the Scientific Revolution*?

- (a) Herbert Butterfield (c) Alexander Koyre
(b) **C.P. Snow** (d) Steven Shapin